Space Force Doctrine Document (SFDD)-1, *The Space Force*

3 April 2025

Overview

- Why We Fight – The Imperative for Spacepower
- Who We Are – Guardians and Military Space Professionals
- What We Do – Employment of Spacepower
- How We Integrate into the Joint Force – Presenting Ready Forces
- History of Military Space Operations
- Military Space Operations Terms of Reference

From the CSO

- This publication codifies why we exist as a Service, who we are as Guardians, and how we employ spacepower to drive the success of the joint force
- To ensure unity of effort, it is essential that each of us fully internalizes these concepts and embodies them as we engage with the entirety of the joint force, our allies, and partners
- Our success or failure hinges entirely upon the performance of Guardians
- Your character, commitment, connection, and courage are why I am confident that the Space Force can fight and win against any adversary

B. CHANCE SALTZMAN
General, USSF
Chief of Space Operations

CSO's Six Undeniable Truths

- Space Force capabilities are critical to the joint force and the American way of life
- The Space Force must defend its capabilities, or the joint force will be unable to project power
- The Space Force must protect the joint / combined force from space-enabled targeting, or the joint / combined force will be unable to meet military objectives
- Space is a warfighting domain, not a collection of supporting activities
- The Space Force is responsible for organizing, training, and equipping critical space capabilities but is also responsible for performance of warfighting operations as an integral part of the joint force
- Guardians are uniquely and specifically trained, educated, and experienced in warfighting activities in, from, and to the space domain

Why We Fight:
The Imperative for Spacepower

What is Spacepower?

- National Spacepower - The totality of a nation's ability to exploit the space domain in pursuit of prosperity and security
  - Securing our national interests in space requires safeguarding and enhancing the ability of US commercial, civil, intelligence, and military entities to safely access, maneuver within, and exploit the space domain without prohibitive interference
- Military Spacepower – The ability to accomplish military objectives in, from, and to the space domain
  - Military spacepower supports and reinforces deterrence by demonstrating America's ability and willingness to impose costs and deny an adversary's benefit from hostile actions

The Service

- The Space Force was established on 20 December 2019 for the purpose of securing the Nation's vital interests in space
  - The Space Force's commitment to the Nation is rooted in the oath every Guardian takes to support and defend the Constitution.
- Title 10 United States Code § 9081(c) directs that the Space Force shall be organized, trained, and equipped to:
  - Provide freedom of operation for the United States in, from, and to space
  - Conduct space operations
  - Protect the interests of the United States in space
- DoDD 5100.01 directs that the Space Force shall develop concepts, doctrine, tactics, techniques, and procedures and organize, train, equip, and provide forces to perform specific functions:
  - Provide freedom of operation for the United States in, from, and to space
  - Provide prompt and sustained space operations
  - Protect the interests of the United States in space
  - Deter aggression in, from, and to space
  - Conduct space operations

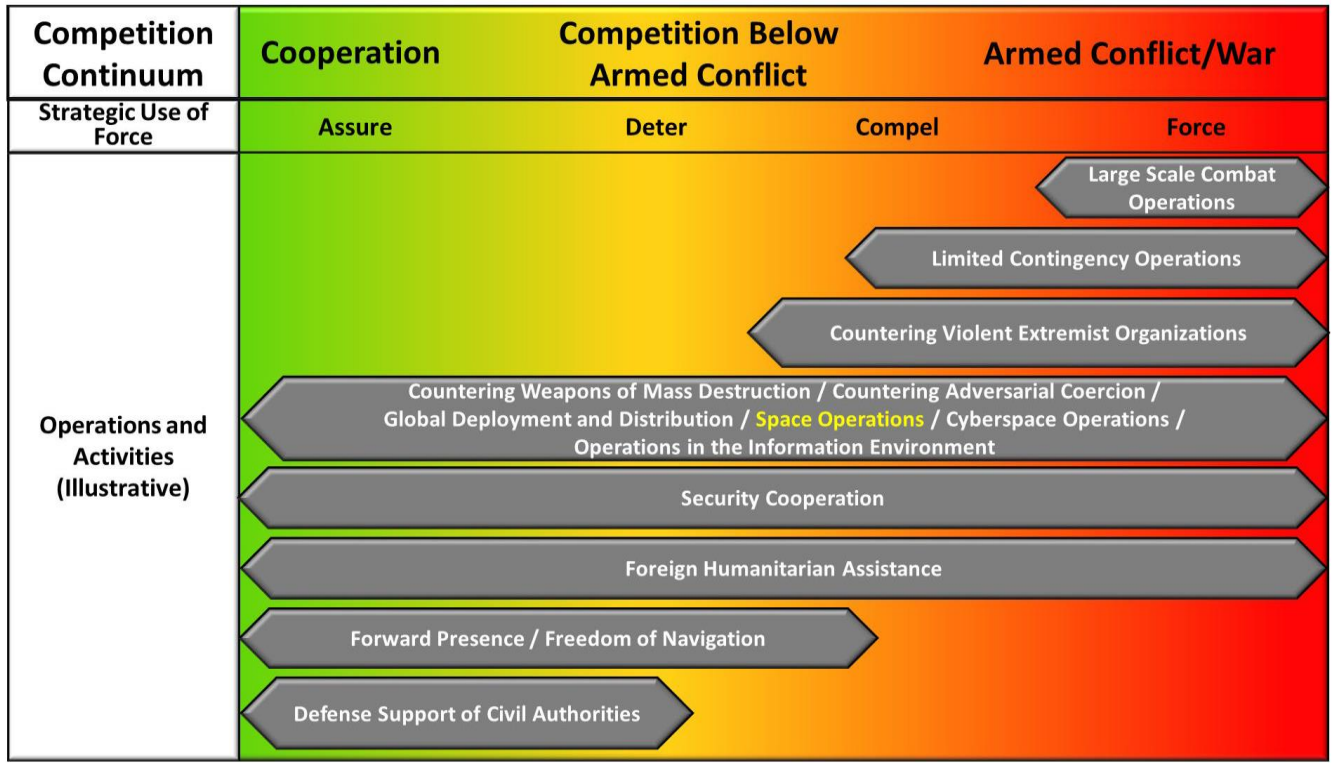
UNITED STATES SPACE FORCE

The Mission

- We perform our mission to:
  - Enable joint force activities
  - Assure our allies and partners
  - Deter, compel, and apply force to our adversaries

The Competition Continuum

- Space forces enable significant effects in **cooperation** and **competition below armed conflict**
  - Offering new avenues of space-related security cooperation to international partners
  - Competing for strategic leadership roles in space
  - Strengthening ties and mutually beneficial relationships with industry partners
  - Bolstering strategic narratives through widely visible space activities



- Military space forces play a key role in **armed conflict and war** through the application of force in the space domain and enabling the application of force in all other domains and environments

Space Warfare

- Space Warfare - To achieve our mission, the Space Force must be prepared to conduct warfare to:
  - Deter or deny attacks on friendly space capabilities by holding adversary space forces at risk
  - Compel an adversary to cease aggressive action in any domain by disrupting, denying, degrading, or destroying the space capabilities they rely on to achieve their military objectives.
  - Undermine an adversary's strategy and their intent to attrit friendly space capabilities through forcible action in, from, and to the space domain.
  - Enable the application of force in all domains by providing a space-enabled combat edge to terrestrial forces.

Competitive Endurance

- Competitive Endurance
  - The Space Force's approach to achieving space superiority, when necessary, in crisis or conflict, while maintaining the safety, security, stability, and long-term sustainability of space
  - Three core tenets:
    - Avoid Operational Surprise
    - Deny First-Mover Advantage
    - Conduct Responsible Counterspace Operations
  - Sets the necessary conditions in peacetime to endure in a long-term state of competition and postures US forces to prevail in conflict

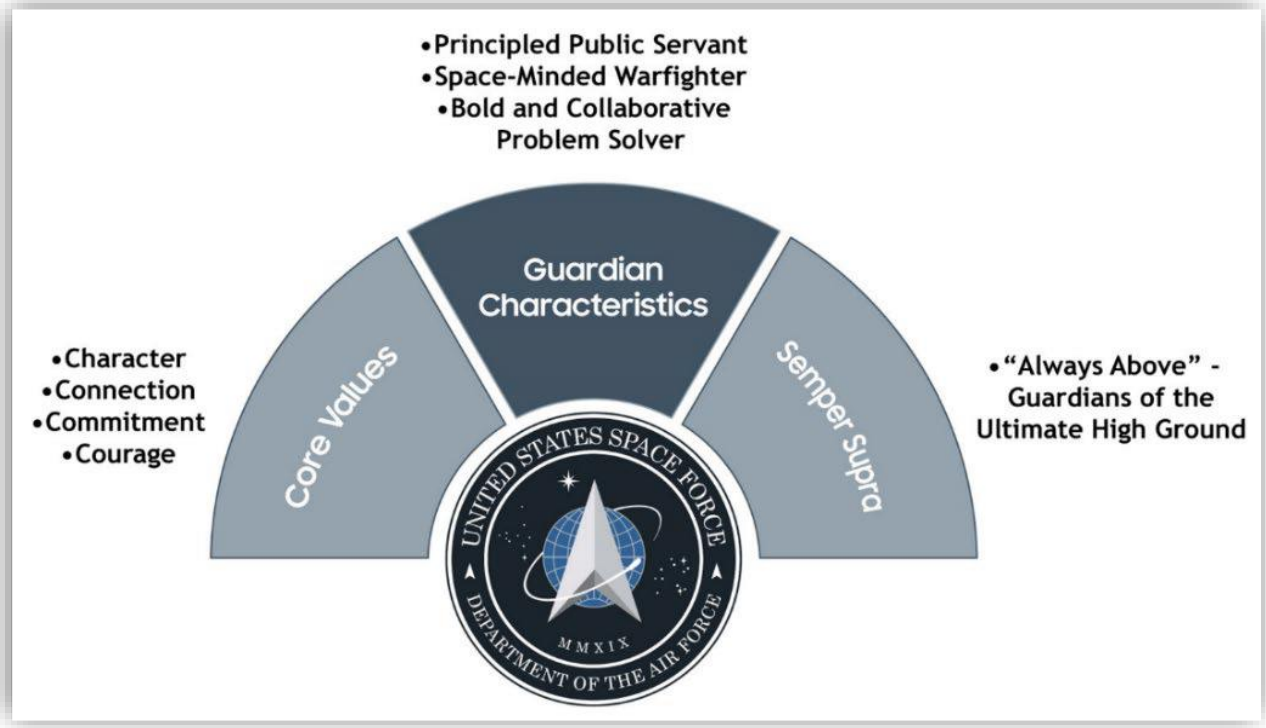
Space Superiority - *The degree of control that allows forces to operate at a time and place of their choosing without prohibitive interference from space or counterspace threats, while also denying the same to an adversary*

Who We Are:
Guardians and Military Space Professionals

Space Force Doctrine Document 1
THE SPACE FORCE
UNITED STATES
SPACE FORCE

Guardians

Guardians



- Our greatest assets are the Guardians who develop, employ, and advance military spacepower for the Nation
- These space-minded warfighters have internalized the Service's mission to secure our Nation's interests in, from, and to space
- The Space Force motto, "Semper Supra," which is Latin for "Always Above," captures the *esprit de corps* of current and future Guardians - securing the ultimate high ground is our duty and promise to the American people

Guardian Values

Guardian Values: Character, Connection, Commitment, and Courage

These values unite us, make us stronger, and sustain us as a ready force of combat-credible warfighters

Guardian Spirit

- The Guardian Spirit embodies the personal and professional characteristics of what it means to be a member of the Space Force
  - Principled Public Servant - Guardians who embody the values of character, connection, commitment, and courage.
  - Space-minded Warfighter - Guardians who are deeply committed to defending our Nation, protecting our interests, and defeating our adversaries.
  - Bold and Collaborative Problem Solver - Guardians who engage with, analyze, and debate new ideas and perpetually challenge the status quo.

Guardian Roles

- Establishes clear roles, responsibilities, and duties for our officers, enlisted, and civilian members
- Codifies a professional development path to meet high-tech demands and build a combat-credible force
  - Officers - Leaders and Planners
    - Expertise across the core functions, planning processes, joint operations and warfare, and command
    - Lead through mission command at all levels
    - Execute planning and operations to integrate space into the joint all-domain fight
  - Enlisted - The Service's Warfighters
    - Frontline operators
    - Extensive operational experience to achieve commander's intent
    - Service's primary source of instructors and innovators
  - Civilians – Corporate Knowledge
    - Continuity across every Space Force function and area of operations
    - Fill functional areas that exist outside of Space Force's officer and enlisted career fields
    - Lead, plan, and manage essential functions through mission command at every level of the Service

Military Service Options

- Full-time and Part-time Service Options
  - More flexibility in the management of our military force
  - Improved military personnel readiness
  - Improved quality of life and retention
  - Attract and retain talent to deliver unmatched space capabilities
  - Eliminates traditional component structure separating active duty from Reserve units

- Guardians in part-time roles
  - Enhance the innovation of the Space Force personnel resource management
  - Support space operations in predictable windows of time
  - Varying part-time durations and compatible with part-time employment

The Space Force Structure

The Space Force Functional Structure

- Streamlined organizational structure supporting globally dispersed Guardians
  - Effectively and efficiently develop, sustain, and employ Space Force capabilities for the joint fight
  - Enable mission command with clear lines of responsibility and authority
  - Standardize where appropriate and remain flexible where needed
- Types of forces
  - Institutional Forces
  - Operational Forces

Operational Forces

- Presented to support operations for the combatant commander (when in the appropriate phase of the Space Force Generation (SPAFORGEN) process)
  - **Employed in Place (EiP) Forces** - Execute their mission from a home station or from an overseas home station, operating as forces presented to a single combatant commander as part of a component field command (C-FLDCOM)
  - **Deployable Space Forces** - Units that physically relocate to support a joint force commander
  - **Service Retained Forces** - Remained assigned to the Service, as reflected in the "Forces For" memorandum and includes Guardians in the Prepare and Ready SPAFORGEN phases

Operational Forces are Discussed Further in Chapter 4

Institutional Forces

- Guardians fulfilling Space Force statutory roles to organize, train, and equip forces
- Non-operational forces that are always under Service control
  - Force Design - Advancing the force by identifying the capabilities, capacity, and force structure required for Guardians to accomplish their roles and missions in the forecasted operational environment
  - Force Development - Preparing the force by adapting and improving our processes, equipment, and thinking to account for projected operational challenges (talent management, education, training, tactics development, test and evaluation, acquisition, and resourcing)

CSO Functions

- Chief of Space Operations
  - Responsible to the Secretary of the Air Force for fulfilling statutory Service roles
  - Member of the Joint Chiefs of Staff
  - Force Design Architect for Space of the Armed Forces
  - Integrator for Department of Defense's Joint Space Requirements

Office of the Chief of Space Operations

Institutional Forces

- Oversees resourcing and programming
- Develops strategy, policy, and guidance for the Service
- Leads requirements generation and force design policy for the Service
- Manages our workforce of the future
- Directs innovation and incorporation of advanced technology
- Enables operations and training to optimize readiness
- Oversees intelligence activities, policies and operations
- Service lead and head:
  - Space Force Intelligence Community Element
  - Defense Intelligence Component
  - Service Cryptologic Component

Field Command & Direct Reporting Units

Institutional Forces

- Align under Headquarters, Space Force
- Execute organize, train, and equip responsibilities on behalf of the Secretary of the Air Force
- Generate, present, sustain, train, and educate Guardians
- Develop, acquire, field, sustain, test, train, and exercise capabilities
- Forecast future threats; develop and validate concepts
- Conduct wargaming activities to prepare the Service

Deltas

Institutional Forces

- Conduct the day-to-day activities to develop and sustain the capabilities, infrastructure, and personnel required for the Service's mission
  - **System Deltas** - Acquisition functions including research, development, and initial fielding of space capabilities
  - **Mission Deltas** - Ensure collaboration between the operations, capability enhancement, and sustainment of a weapon system with the addition of cybersecurity and intelligence support to ensure readiness
  - **Space Launch Deltas** - Provide agile, responsive, and resilient spaceports, launch and test range capabilities
  - **Space Base Deltas** - Provide installation operations support and community support services

See Chapter 4 for Combat Deltas – Operational Forces

Subordinate Units

Institutional Forces

- **Squadrons** -Perform the preponderance of mission execution under the direction of their parent Delta
  - Mission squadrons
  - Combat training squadrons
  - Space range squadrons
  - Space aggressor squadrons
  - Force development squadrons

- **Detachments** - May form within an existing squadron due to mission requirements, often function as geographically separated units

See Chapter 4 for Combat Squadrons and Detachments – Operational Forces

Relationships

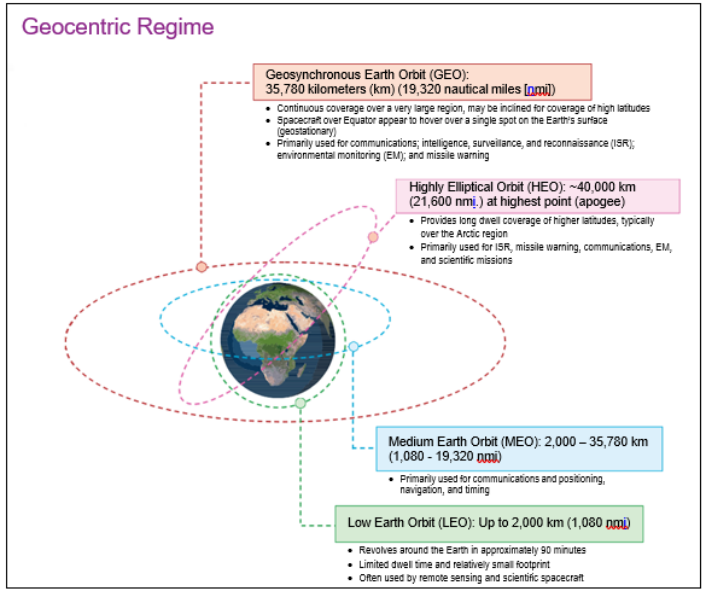
- Air Force, sister Services, other US government agencies, allies, partners, commercial providers, and academia
  - Share valuable data and resources
  - Conduct mutually beneficial activities within a shared operational environment
  - Capitalize on new technology and perspectives
  - Leverage shared capabilities that enhance space operations
  - Complicate adversary decisions
  - Creating opportunities to:
    - Integrate and synchronize space operations
    - Produce enduring operational advantages
    - Reinforce global norms of responsible behavior
    - Benefit the United States, and our allies and international partners

What We Do:
Employment of Spacepower

Operating In, From, and To the Space Domain

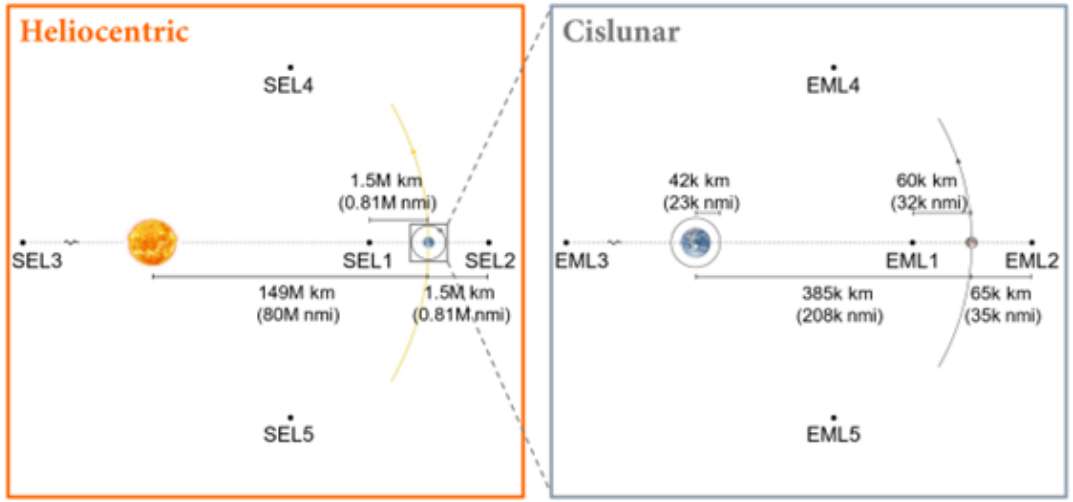
- The area above the altitude where atmospheric effects on airborne objects become negligible
- An austere, highly radiated environment congested by environmental and human-made debris and subject to drastic temperature fluctuations
- Physics in the domain create challenges for position, movement, maneuver, security, and safety of flight

Orbital Regimes



- The Space Force currently considers three orbital regimes for current and future space operation
  - Geocentric Regime - The spherical region of space where Earth's gravity dominates, and objects follow orbits and trajectories around the Earth, extending out to and beyond geosynchronous Earth orbits (including but not limited to transfer orbits and the disposal belt).
  - Cislunar Regime – The spherical region extending out from geosynchronous orbit to the Second Earth-Moon Lagrange point and characterized by the combined gravitational effects of the Earth and/or the Moon. This includes the space between these bodies, the orbits and trajectories around the Earth-Moon Lagrange points, and lunar orbits where the Moon's gravity is dominant
  - Heliocentric Regime - Defined by the Sun's gravitational field, encompasses regions in the solar system that are not primarily under the influence of the Earth, Moon, or other solar system body. The heliocentric regime also includes the Sun-Earth Lagrange points and their associated orbits and trajectories

An orbital regime is a region in space associated with a dominant gravitational system capable of capturing the orbit of other objects



Other Domains & Environments

- Operations in the space domain support and are supported by operations in other domains and environments
  - Electromagnetic Environment
    - The electromagnetic spectrum is essential to all space operations, enabling communications, sensing, and the delivery of information and effects via transmitted signals in an increasingly congested and contested environment
  - Cyberspace
    - All space operations are conducting within cyberspace and share the environment with allied, neutral, and adversarial parties
    - The ability to project spacepower relies on the ability to maneuver to, from, and through cyberspace
  - Air, Land, and Maritime Domains
    - These domains possess critical infrastructure upon which the success of friendly and adversarial space operations rely
    - They also host the user equipment which receives space-enabled information
    - Awareness and protection of activities, infrastructure, and personnel in these domains are critical to military space operations

Space Force Core & Enterprise Functions

- Describe the operational and support activities a military force – either red or blue – could perform in, from, or to the space domain
- Informed by existing technology rather than policy or resourcing, the inclusion of a mission area does not imply the existence of a corresponding capability

- **Core Functions:** Our core functions are the broad and enduring operational roles that we organize, train, and equip Guardians to perform

- **Enterprise Functions:** Four cross-cutting enterprise functions which are key enablers of the core functions

Space Force Core & Enterprise Functions

Core Functions

Space Control

Activities required to contest and control the space domain.

Orbital Warfare
- Offensive Orbital Warfare
- Defensive Orbital Warfare

Electromagnetic Warfare
- Offensive Space Control Electromagnetic Warfare
- Defensive Space Control Electromagnetic Warfare

Cyberspace Warfare
- Offensive Cyberspace Warfare
- Defensive Cyberspace Warfare

Global Mission Operations

Activities to deliver space capabilities to the Joint Force and the Nation.

Satellite Communication (SATCOM)
- Strategic SATCOM
- Operational SATCOM
- Tactical SATCOM

Navigation Warfare

Missile Warning and Tracking
- Ground-Based Surveillance
- Overhead Persistent Infrared Surveillance

Space-Based Sensing and Targeting
- Reconnaissance
- Surveillance
- Target Acquisition

Theater Electromagnetic Warfare
- Electromagnetic Attack
- Electromagnetic Protection
- Electromagnetic Surveillance

Space Access

The movement and sustainment of equipment in, from, and to the space domain.

Satellite Control

Spacelift
- Launch
- Recovery
- Payload Processing
- Telemetry and Tracking
- Responsive Launch

Launch Range Control
- Restricted Area Control
- Flight Safety

Enterprise Functions

Enterprise

Intelligence
- Collection
- Processing and Exploitation
- Analysis and Production
- Integration
- Targeting

Cyberspace Operations
- Information Network Operations
- Defensive Cyberspace Operations

Command and Control (C2)
- Global C2
- Regional C2
- Tactical C2
- Battle Management

Space Domain Awareness
- Reconnaissance
- Surveillance
- Conjunction Assessment
- Collision on Launch Assessment
- Breakups
- Rendezvous and Proximity Operations Support

Core Function: Space Control

Space Control

Activities required to contest and control the space domain.

Orbital Warfare
- Offensive Orbital Warfare
- Defensive Orbital Warfare

Electromagnetic Warfare
- Offensive Space Control Electromagnetic Warfare
- Defensive Space Control Electromagnetic Warfare

Cyberspace Warfare
- Offensive Cyberspace Warfare
- Defensive Cyberspace Warfare

- **Space Control** - Activities required to contest and control the space domain
  - **Orbital Warfare** - Combat operations conducted through fires, movement, and maneuver to control the space domain
  - **Electromagnetic Warfare** - Combat operations on the link segment through electromagnetic spectrum fires to control the space domain
  - **Cyberspace Warfare** - Combat operations conducted in cyberspace through fires, movement, and maneuver to control the space domain

Core Function: Global Mission Operations

- **Global Mission Operations** - Activities to deliver space capabilities to the joint force and the Nation
  - **Satellite Communication** - Secure, resilient communication services provided through space-based platforms
  - **Navigation Warfare** - The assurance or denial of PNT data through space, cyberspace, and electromagnetic spectrum operations
  - **Missile Warning and Tracking** - Activities to detect missile launches, track their trajectory, and issue timely warnings
  - **Space-Based Sensing and Targeting** - Space-based activities and operations to provide terrestrial battlespace awareness
  - **Theater Electromagnetic Warfare** - Actions taken in the electromagnetic spectrum to protect or prevent the ability to communicate using space-based platforms

Global Mission Operations

Activities to deliver space capabilities to the Joint Force and the Nation.

Satellite Communication (SATCOM)
- Strategic SATCOM
- Operational SATCOM
- Tactical SATCOM

Navigation Warfare

Missile Warning and Tracking
- Ground-Based Surveillance
- Overhead Persistent Infrared Surveillance

Space-Based Sensing and Targeting
- Reconnaissance
- Surveillance
- Target Acquisition

Theater Electromagnetic Warfare
- Electromagnetic Attack
- Electromagnetic Protection
- Electromagnetic Surveillance

Core Function: Space Access

Space Access

The movement and sustainment of equipment in, from, and to the space domain.

Satellite Control

Spacelift
- Launch
- Recovery
- Payload Processing
- Telemetry and Tracking
- Responsive Launch

Launch Range Control
- Restricted Area Control
- Flight Safety

- **Space Access** - The movement and sustainment of equipment in, from, and to the space domain
  - **Satellite Control** - Activities necessary to assure the infrastructure, networks, equipment, and connectivity that enables mission control (i.e., telemetry, tracking, and commanding) of satellites
  - **Spacelift** - The movement of payloads (spacecraft or other materials) to and from the space domain
  - **Launch Range Control** - Activities to ensure the provision of a safe and secure range for space launch operations

Enterprise Functions

- **Intelligence** - The products resulting from the collection, processing and exploitation, analysis and production, integration, and targeting concerning foreign nations, hostile or potentially hostile forces, and areas of actual or potential operations
- **Cyberspace Operations** - The employment of networked, stand-alone, and platform-embedded information technology infrastructure for the access to and use of data to achieve effects in or through cyberspace
- **Command and Control** - The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of the mission
- **Space Domain Awareness** - The detection, characterization, attribution, prediction, and targeting of activities in the space domain to inform decision making

Space Systems Architecture

- Guardians employ space systems to conduct activities and create effects in, from, and to the space domain
- Space systems include components in three segments operating across all operational environments and domains
  - Orbital Segment - The orbital segment includes space systems operating in the environment of the space domain
  - Terrestrial Segment- The terrestrial segment systems operate in the land, air, and maritime domains.
  - Link Segment - The link segment of space systems operates in the electromagnetic spectrum.
- The integration of the space architecture across the three segments determines capabilities, limitations, and vulnerabilities for space operations
- Preserving a space capability requires protection in all three segments
- An attack on any segment of the space architecture can deny, degrade, disrupt, or destroy a space capability

Command and Control

- C2 is the exercise of authority and direction by a properly designated commander over assigned, attached, or allocated forces in the accomplishment of the mission
- Includes three basic functions:
  - Gaining and maintaining situational awareness
  - Enabling operational decision making
  - Directing forces
- Command
  - Includes both the authority and responsibility to use resources to accomplish assigned missions
  - Requires motivating and directing people and organizations to accomplish missions.
- Control
  - Inherent in command
  - Allows commanders to manage and direct forces and functions consistent with their authority
  - Helps commanders and staff compute requirements, allocate means, and integrate efforts
  - Necessary to determine the status of organizational effectiveness, identify variance from set standards, and correct deviations from these standards

Command and Control

- C2 of space forces reflects the distinctive character of space operations and the unique attributes of the space domain
  - Capabilities support multiple AORs simultaneously, others are rapidly shifted, and some cannot persistently support a single AOR
  - Space forces may operate completely within one AOR to support another combatant command
  - Operations in, from, and to the space domain regularly cross AOR boundaries
- C2 of spacepower in any AOR requires coordinating, deconflicting, and as necessary, controlling the transit of effects in and out of other combatant commands' AORs
- Effective command and control is essential to the application of spacepower

Guiding Principles for the C2 of Space Forces

- **Commander Driven** - Commanders have command authority by virtue of rank or assignment and are expected to exercise their responsibility for C2 and staffs and operations centers execute these authorities on behalf of the commander
- **Command at Echelon** - The Space Force organizes combat forces with at least two echelons of command: (1) Service component and (2) combat formation
- **Mission Command** - The most effective way to C2 space forces in a contested, degraded, and operationally limited environment is through C2 at echelon using the principles of mission command
- **Shared Understanding** - Commanders and their staffs require relevant information to facilitate sound and timely decision-making, key to successful military operations at all echelons
- **Process Focused** - C2 is a continuous process which can be made more effective by tailored materiel systems and a consistent battle rhythm commensurate with a Service's warfighting concepts
- **Integration and Synchronization** - Integration and synchronization with other Service and functional components, interagency mission partners, commercial providers, and coalition partners occurs at echelon level and across domains
- **Decentralized Execution** - Effective C2 at echelon requires commanders to delegate planning and execution responsibilities to the lowest appropriate level
- **Coordination** - Effective C2 requires direct liaison authority and robust coordination agreements so that combat formations can directly and effectively coordinate with each other during execution without additional higher echelon guidance

Mission Command

- Leadership philosophy centered on the art of command, the science of control, and the discipline of decision making.
  - Empowers Guardians to exercise sound judgment in how they conduct their assigned tasks
  - Emphasizes mutual trust, disciplined initiative, and innovation
  - Applies to decision making across all Space Force organizations
  - Requires continuous effort to understand, adapt, and effectively direct the achievement of intent under evolving conditions and environments
- Characterizes the Space Force's approach to C2
  - Empowers Guardians to make decisions at the necessary operations tempo, enabled by mission-type orders, conditional authorities, and control and coordination procedures
  - Increases the agility and flexibility of operations in, from, and to the space domain

Battle Management

- Management of activities within the operational environment based on the commands, direction, and guidance given by appropriate authority
  - Inherently multi-domain and demands joint and multi-disciplinary competencies
  - Includes direction, coordination, integration, and deconfliction of fires, maneuvers, information, links, and sensors during operational execution
- A collection of enduring assumptions apply to battle management in every domain
  - No two entities in the battlespace are in identical states simultaneously
  - Current, credible information on the disposition of friendly, hostile, and other forces is essential
  - The distribution of entities in the battlespace is not uniform
  - The battlespace evolves over time
  - Information available to decision makers is incomplete and may be inaccurate
  - The forces available for a mission are not infinite in quantity, availability, and readiness

How We Integrate into the Joint Force: Presenting Ready Forces

Presentation of Forces

- As a Service Force Provider - the Space Force prepares and presents forces to the combatant commanders

- Presented Space Force forces
  - Control, plan, direct, and execute operations under the authority of a joint force commander
  - Fulfill strategic, operational, and tactical objectives through the application of spacepower

Component Field Commands

- C-FLDCOM is the organizational structure through which a Commander, Space Force Forces (COMSPACEFOR) conducts Service component responsibilities
  - C2 of assigned, attached, or allocated space forces
  - Coordinate space operations to support the combatant commander's campaign plan
  - Synchronize space effects with the other Service components
  - Integrate allied and partner space capabilities as force multipliers
  - Support national, civil, international, and commercial users as appropriate
  - Conduct space-related security cooperation activities with allies and partners

- Each C-FLDCOM is comprised of Guardians from multiple career fields performing operations under the combatant command authority of the combatant commander
  - A COMSPACEFOR controls these Space Force formations and operations
  - These units are responsible for planning, directing, and executing operations, and assisting with collaboration between Service component commands and the combatant command

Component Field Commands

Through component field commands, the Space Force coordinates and integrates space forces into planning and current operations within the combatant commands.

Commander of Space Force Forces (COMSPACEFOR)

- A COMSPACEFOR leads each of the Service components (C-FLDCOMs) presented by the Space Force to combatant commands

- Administrative Control (ADCON)
  - Direction or exercise of authority over subordinate or other organizations in respect to administration and support
    - COMSPACEFOR exercises administrative control over assigned or attached Space Force forces, including responsibility for oversight and discipline of Guardians and members within the C-FLDCOM
    - ADCON of USSPACECOM-presented Space Force combat forces passes to Commander, Space Forces Space (S4S/CC)

Commander of Space Force Forces (COMSPACEFOR)

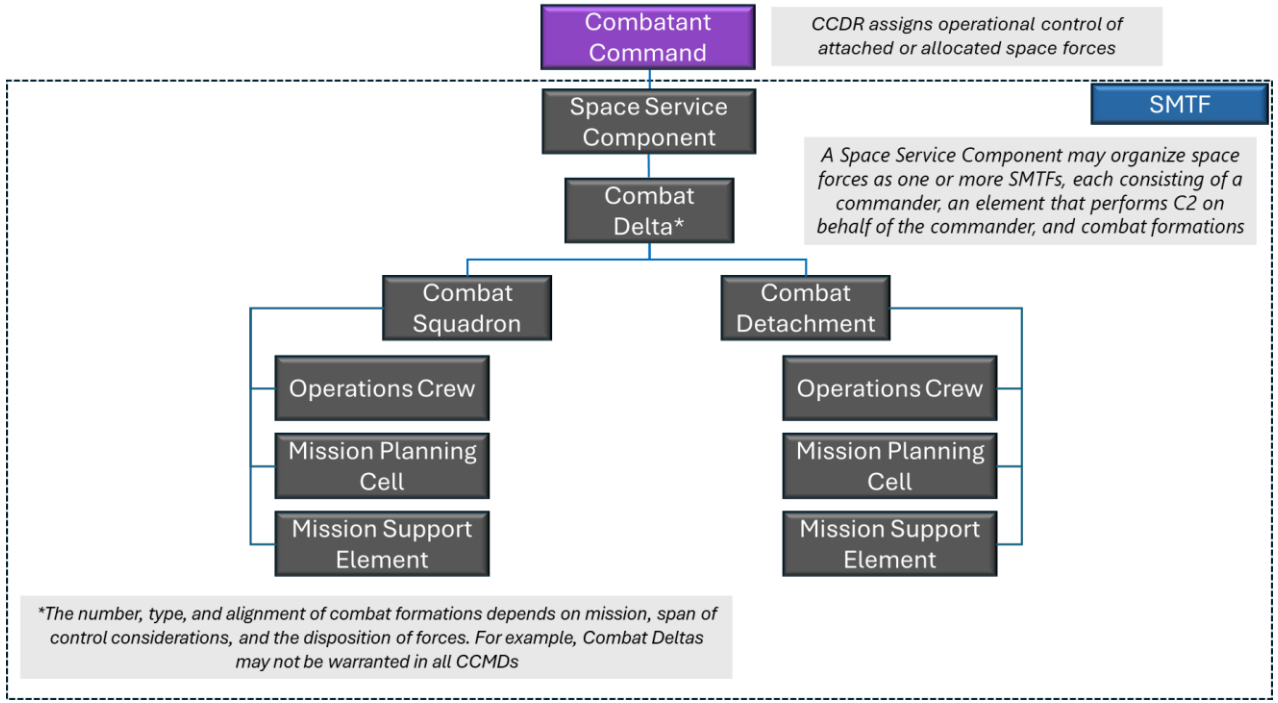
- Operational Control (OPCON)
  - The authority to perform those functions of command over subordinate forces involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction necessary to accomplish the mission
    - Combatant commander, or the joint force commander may delegate OPCON of assigned, attached, or allocated space forces to the Service component commander with the preponderance of forces and the ability to C2 those forces
    - OPCON of assigned, attached, or allocated space forces is delegated to the COMSPACEFOR when assigned as Combined/Joint Forces Space Component Commander (C/JFSCC) to allow him/her to organize available forces and further delegate applicable authorities
- Tactical Control (TACON)
  - The authority over forces that is limited to the detailed direction and control of movements or maneuvers within the operational area necessary to accomplish missions or tasks assigned.
    - If designated as the C/JFSCC, the COMSPACEFOR assigns tasks to subordinate commanders whether they are Space Force combat detachments, units from another Service, or coalition partners to accomplish the mission

Space Mission Task Force (SMTF)

- Warfighting structure for space operations tasked to execute a named operation of a combatant command

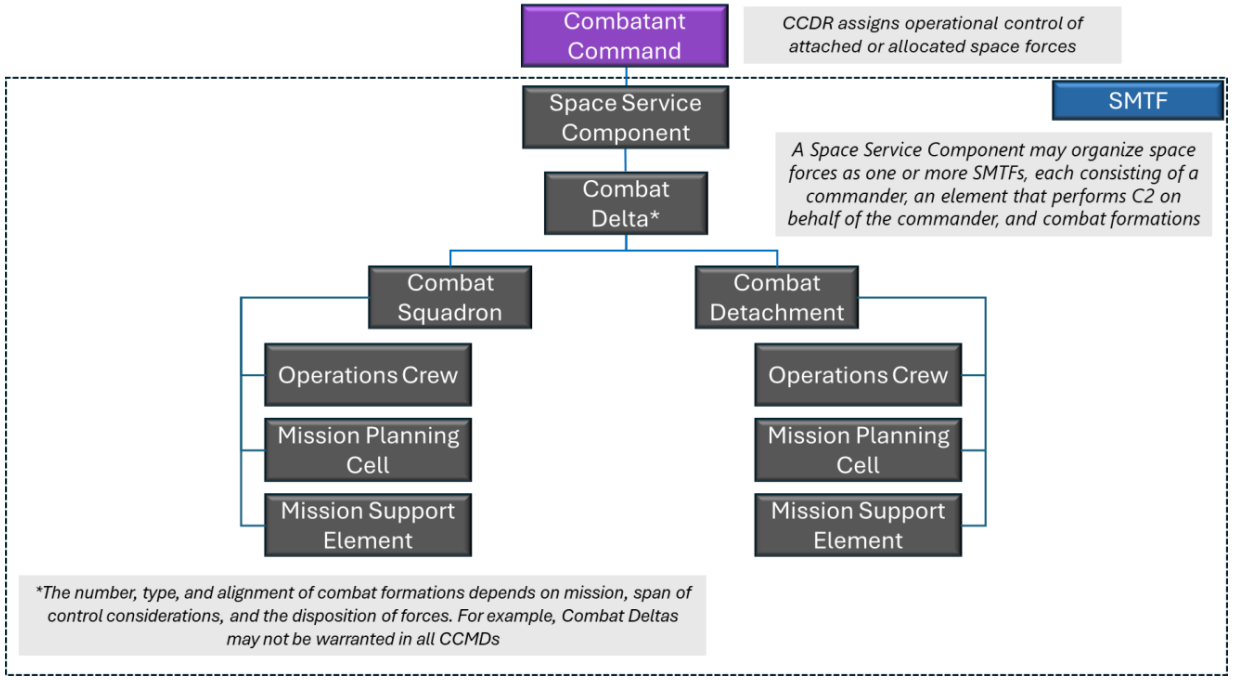
- Aggregation of elements including:
  - Commander
  - C2 element
  - Combat formations

- COMSPACEFOR may organize subordinate space forces as one or more SMTF
  - When a combatant commander gives operational control to a COMSPACEFOR of assigned, attached, or allocated space forces
  - Mission or span of control requirements dictate
  - Structure varies based on span of control considerations and the number of forces in theater



Combat Formations

- Types of combat formations:
  - Combat Deltas
  - Combat Squadrons
  - Combat Detachments
  - Force Elements

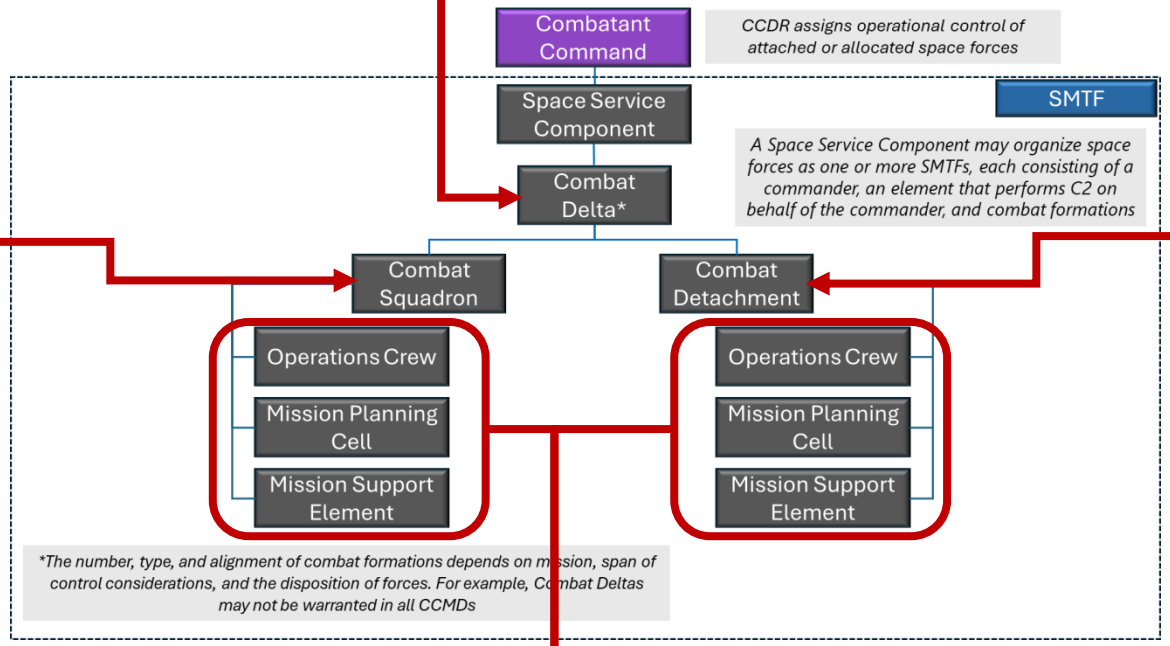


- Presented to the combatant commanders via a C-FLDCOM
- Force elements from one or more of the force generation squadrons will form these combat formations

Organizational Structure of Presented Forces

Combat Deltas - The Space Force presents combat deltas to a combatant commander through the C-FLDCOM. Combat deltas support the operational-level mission space operations planning, execute C2 of assigned, attached, or allocated forces, and synchronize and coordinate with allies and partners operating at a commensurate level. If directed, these units coordinate space requirements with other combatant commands, agencies, allies, and partners

Combat Squadron – An employed-in-place operational formation assigned to a combatant command to provide space effects supporting combatant-command mission requirements



Combat Detachments - The Space Force's deployable operational formation. Detachments deploy as forces allocated to support of combatant-command requirements

Force Elements - Subordinate formation of a combat squadron or combat detachment that performs a distinct activity in support of the mission

- **Operations Crew** - Performs weapon system operations in support of mission activities
- **Mission Planning Cell** - Leads planning, scheduling, and assessment for operational activities
- **Mission Support Element** - Conducts support activities like engineering, analysis, maintenance, and communications. These may include Airmen in non-Space Force specialties who provide capabilities required to execute the mission (e.g., force protection, emergency services, civil engineering)

Space Force Generation (SPAFORGEN)

- The DoD requires every Service to monitor and assess their operational readiness and capabilities for forces assigned, attached, or allocated to combatant commands.
- SPAFORGEN is the Space Force's model to manage the battle rhythm of rotational forces, ensuring these forces can maintain the highest possible level of overall readiness.
- All force generation units follow the three-phases of the SPAFORGEN model.
  - Prepare Phase - Personnel reconstitution, leave, and replenishment of supplies and equipment. Force elements also conduct Service-led activities such as training and positional upgrades, professional military education, and qualification training
  - Ready Phase - Training, small and large force-employment exercises, squadron-level validations, and other preparations for operations in a contested space domain against a thinking adversary
  - Commit Phase - Fully resourced, validated, and ready to conduct operations and are assigned, attached, or allocated to a combatant command for an operational mission
- The relative length of each phase varies for combat squadrons and combat detachments due to the nature of mission requirements for employed-in-place versus deployed operations
- SPAFORGEN establishes and maintains a predictable, standardized battle rhythm ensuring rotational forces

History of Military Space Operations

- The roots of US military space operations are traceable back to rocket and spacecraft feasibility studies in the immediate post-World War II period, but not until the 1950s did actual orbital operations become a reality
- November 1963, the Air Force satellite control facility in Sunnyvale was “nationalized” to serve as the nucleus for development of all DoD space programs
- Air Force Systems Command managed a considerable portion of military space activities, other organizations gained relevant responsibilities as space matured
- In April 1982 Air Force Chief of Staff General Lew Allen directed General James Hartinger to organize an Air Force major command for space
  - Air Force Space Command (AFSPC) activated in September 1982
- United States Space Command (USSPACECOM) was activated in September 1985 under General Robert Herre
  - USSPACECOM employed the full range of space capabilities in theater combat operations for the first time during Operations Desert Shield and Desert Storm
  - Over the next decade USSPACECOM further operationalized and normalized space capabilities to support US domination of the battlespace
- Terrorist attacks on 11 September 2001 compelled senior military leaders to establish a United States Northern Command and disestablish USSPACECOM, operational responsibilities for space merged into United States Strategic Command (USSTRATCOM) on 1 October 2002
  - USSTRATCOM formed Joint Functional Component Command, Space (JFCC-Space) in July 2006 commanded by dual-hatted 14th Air Force Commander
  - Joint Space Operations Center opened at Vandenberg Air Force Base in May 2005 – redesignated Combined Space Operations Center in July 2018 reflecting the presence of allies and partners
- Integration of space and intelligence led to the establishment of the Joint Interagency Combined Space Operations Center at Schriever Air Force Base in October 2016; it became the National Space Defense Center in April 2017
- On 1 December 2017 USSTRATCOM Commander, General John Hyten and AFSPC Commander, General John Raymond normalized Service component alignment by reassigning the JFCC-Space commander’s responsibilities to the AFSPC commander, transforming AFSPC into a warfighting headquarters
- On 29 August 2019 USSPACECOM was officially re-established
- Presidential signing of the Fiscal Year 2020 National Defense Authorization Act on 20 December 2019 officially redesignated Headquarters AFSPC as Headquarters United States Space Force and created the CSO as its leader
- General John Raymond was the first CSO and dual-hatted as CDRUSSPACECOM

Military Space Operations Terms of Reference

- Organizes and describes the operational and support activities a military force – either red or blue – could perform in, from, or to the space domain
- The MSO ToR is informed by existing technology rather than policy or resourcing, the inclusion of a mission area does not imply the existence of a corresponding capability

- Framework
  - Core Function - A broad and enduring operational role a Service has designed, equipped, and trained its forces to perform
    - Mission Area - A major grouping of interrelated activities required to accomplish a core function
    - Mission Set - A family of capabilities that performs similar tactical tasks
  - Enterprise Function - A cross-cutting activity that enables all core functions
    - Enterprise Activity - An element or subdivision of an enterprise function

Space Control

Activities required to contest and control the space domain.

Orbital Warfare
- Offensive Orbital Warfare
- Defensive Orbital Warfare

Electromagnetic Warfare
- Offensive Space Control Electromagnetic Warfare
- Defensive Space Control Electromagnetic Warfare

Cyberspace Warfare
- Offensive Cyberspace Warfare
- Defensive Cyberspace Warfare

Global Mission Operations

Activities to deliver space capabilities to the Joint Force and the Nation.

Satellite Communication (SATCOM)
- Strategic SATCOM
- Operational SATCOM
- Tactical SATCOM

Navigation Warfare

Missile Warning and Tracking
- Ground-Based Surveillance
- Overhead Persistent Infrared Surveillance

Space-Based Sensing and Targeting
- Reconnaissance
- Surveillance
- Target Acquisition

Theater Electromagnetic Warfare
- Electromagnetic Attack
- Electromagnetic Protection
- Electromagnetic Surveillance

Space Access

The movement and sustainment of equipment in, from, and to the space domain.

Satellite Control

Spacelift
- Launch
- Recovery
- Payload Processing
- Telemetry and Tracking
- Responsive Launch

Launch Range Control
- Restricted Area Control
- Flight Safety

Enterprise

Intelligence
- Collection
- Processing and Exploitation
- Analysis and Production
- Integration
- Targeting

Cyberspace Operations
- Information Network Operations
- Defensive Cyberspace Operations

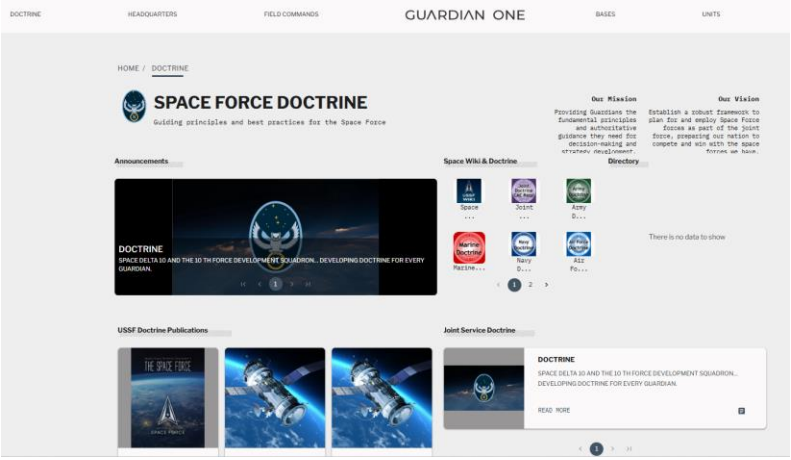
Command and Control (C2)
- Global C2
- Regional C2
- Tactical C2
- Battle Management

Space Domain Awareness
- Reconnaissance
- Surveillance
- Conjunction Assessment
- Collision on Launch Assessment
- Breakups
- Rendezvous and Proximity Operations Support

Accessing Space Force Doctrine

Space Force Portal

https://my.spaceforce.mil/landing-page/doctrine

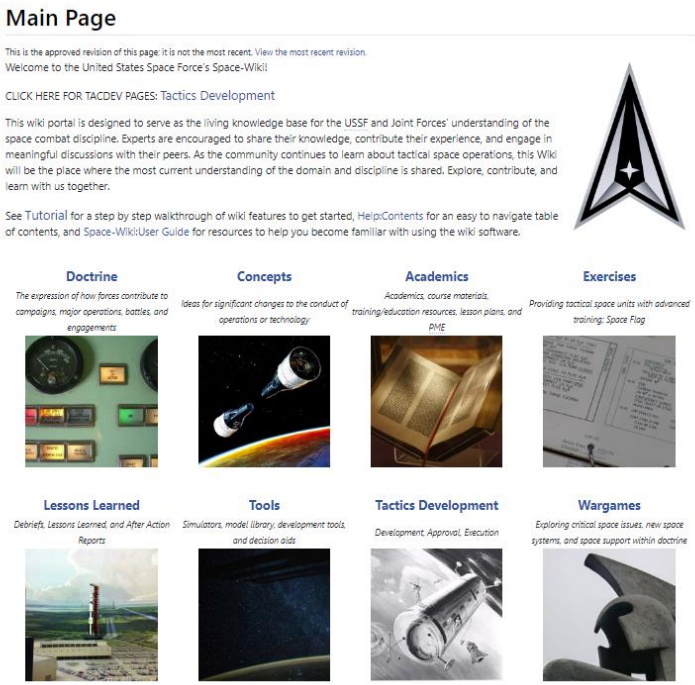


STARCOM Digital Library

https://www.starcom.spaceforce.mil/Resources/Digital-Library/

Space Wiki

NIPR: https://space-wiki.apps.dso.mil/wiki/Main_Page



SIPR: https://space-wiki.mission-prod.odin.dso.smil.mil

JWICS: https://space-wiki.mission-prod.odin.dso.ic.gov

Guardian One App

Space Delta 10
Space Training and Readiness Command

Wargaming | Field Experimentation |Doctrine | Lessons Learned